CIE

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TITLE: Design and Development of a Real-Time Collaborative Code Editing Platform

SCHOOL: Chitkara University, Rajpura, Punjab

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ABSTRACT

1.1 Problem Background

In recent years, collaborative software development has become increasingly important due to the rise of remote work, online learning, and distributed development teams. Traditional code editors and development environments often lack real-time collaboration capabilities, requiring developers to rely on external communication tools to share code, resolve issues, and coordinate tasks. This leads to delays, version conflicts, and reduced productivity, especially in academic environments, hackathons, and team-based projects. There is a growing need for an integrated platform that allows multiple users to write, edit, and review code simultaneously in a shared environment.

1.2 Objective of the Project

The objective of this project is to design and develop a real-time collaborative code editor that enables multiple users to write and edit source code simultaneously, share cursor positions, and collaborate efficiently while maintaining synchronization, usability, and secure access.

1.3 Proposed Solution / System Overview

The proposed system, named **CodeSphere**, is a web-based real-time collaborative code editor designed to support synchronized multi-user programming. The system functions by allowing users to join a shared coding room using a unique identifier, where all code changes are reflected instantly for every participant. The major components of the system include a real-time communication server, a collaborative code editor module, user authentication and session management, and synchronization services. These components work together to ensure consistent code updates, secure collaboration, and support copyright documentation and ownership clarity by preserving originality and controlled access.

1.4 Key Features or Functionality

Key features of the system include real-time code editing, multi-user collaboration, cursor position tracking, language selection support, code formatting, and file download functionality. The system also provides room-based collaboration, live user presence display, and automatic synchronization for newly joined users. These features enhance teamwork, reduce communication gaps, and improve coding efficiency in collaborative environments.

1.5 Technology / Method Used

The system is developed using modern web technologies such as React for the frontend interface, Node.js and Express for backend services, and Socket.IO for real-time communication. CodeMirror is used as the core code editor to support syntax highlighting and multiple programming languages.

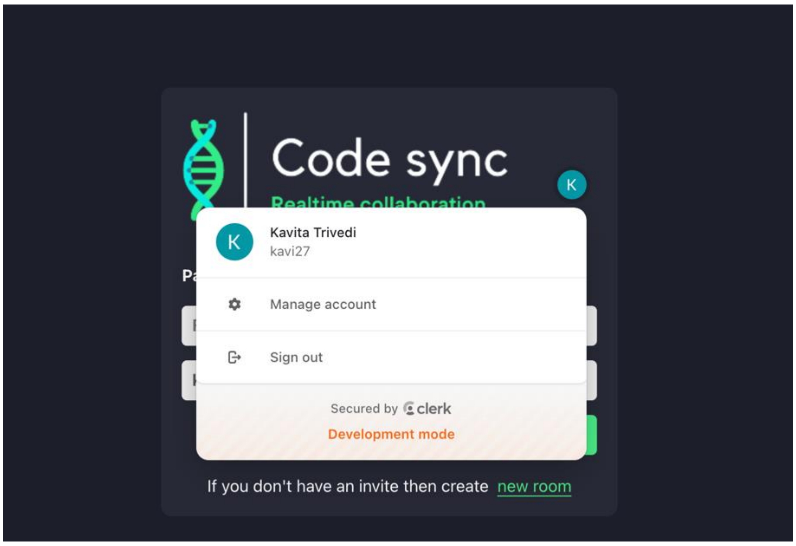
1.6 Expected Outcome & Benefits

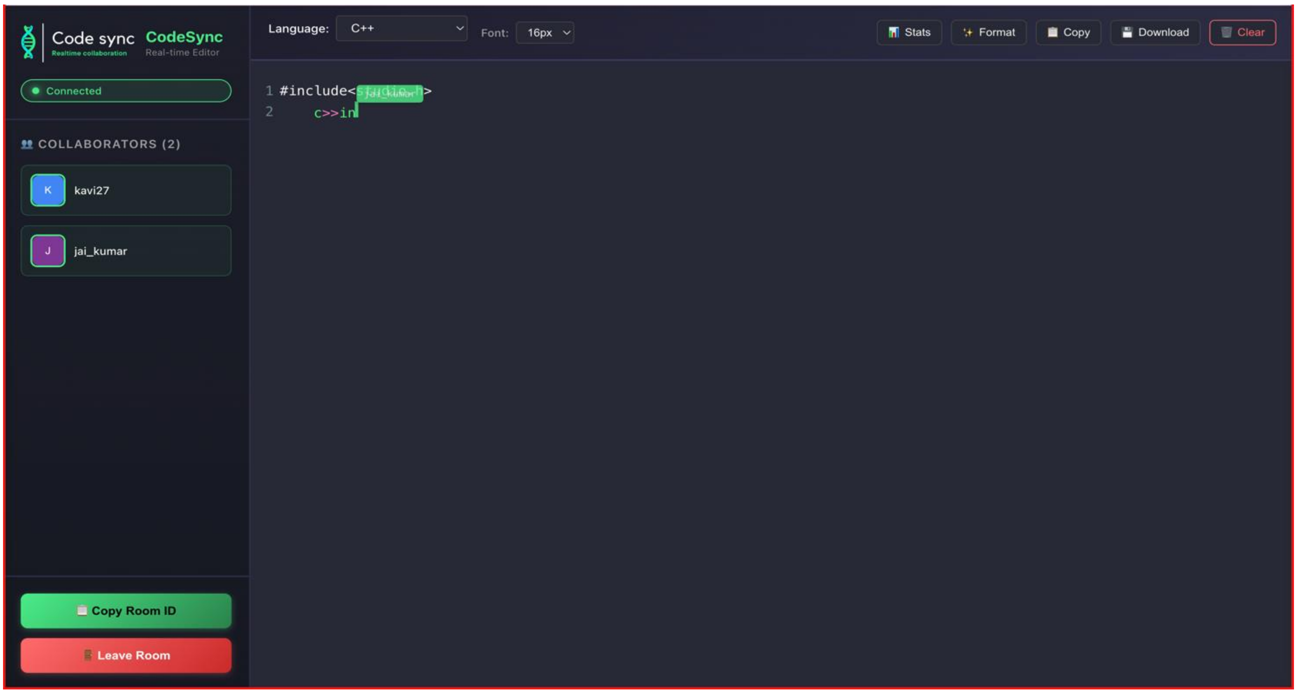
The expected outcome of this project is a fully functional real-time collaborative coding platform that improves productivity, learning efficiency, and teamwork. The system reduces dependency on external communication tools, minimizes code conflicts, and enables seamless collaboration. It is particularly beneficial for students, developers, and educators involved in remote coding sessions, group projects, and technical interviews.

1.7 Original Contribution Statement

This project represents an original software development effort by the author, integrating real-time collaboration features into a single unified coding platform without adapting or copying any existing proprietary system.

1.8 Project Screenshots




Code sync
Realtime collaboration
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Real-time Editor
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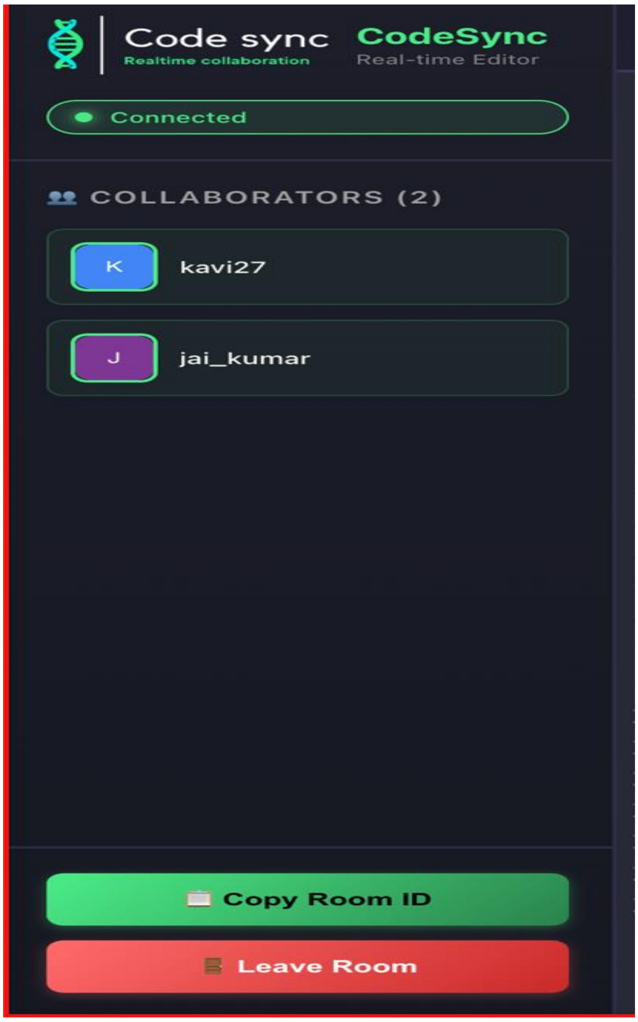
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Realtime collaboration
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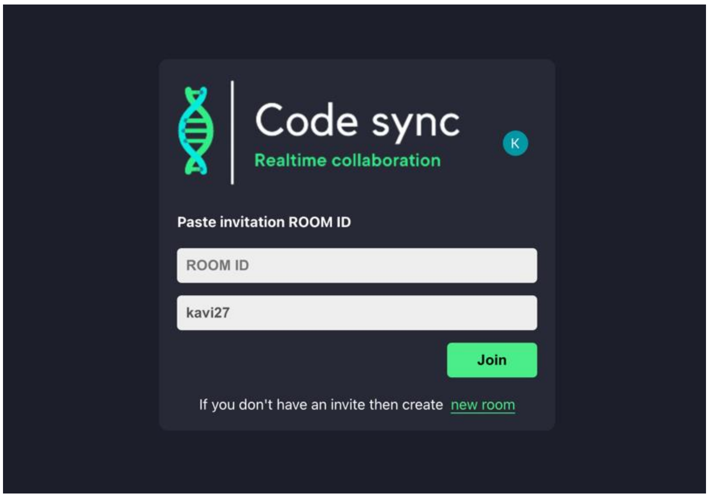
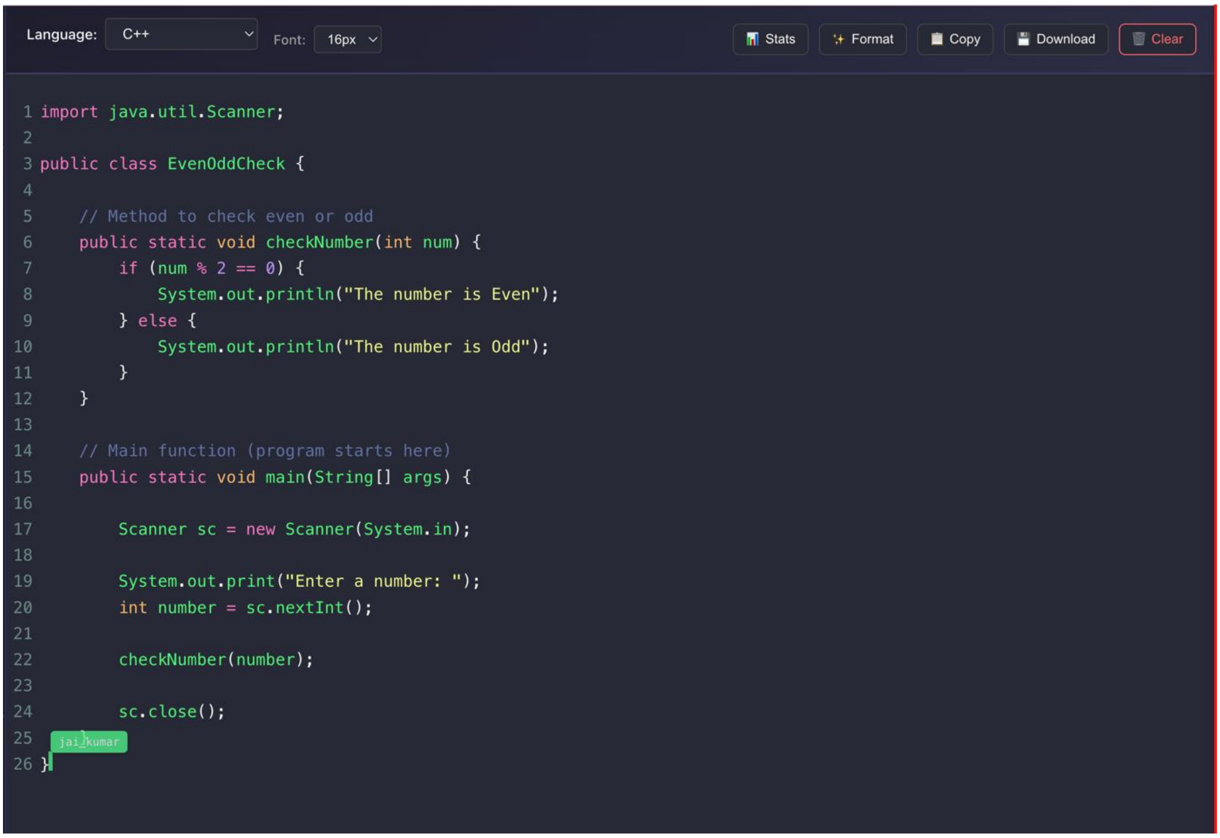

Code sync
Realtime collaboration
K
Paste invitation ROOM ID
ROOM ID
kavi27
Join
If you don't have an invite then create new room


Language:
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Clear
import java.util.Scanner;
public class EvenOddCheck {
// Method to check even or odd
public static void checkNumber(int num) {
if (num % 2 == 0) {
System.out.println("The number is Even");
} else {
System.out.println("The number is Odd");
}
}
// Main function (program starts here)
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("Enter a number: ");
int number = sc.nextInt();
checkNumber(number);
sc.close();
}
jai_kumar
}